

Naive Bayes Algorithm.

Bayes Theorem:

↳ Bayes theorem is widely used for text classification.

↳ Naive Bayes was the first algorithm to be designed for Email spam filter.

If A and B any ~~two~~ events,

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Posterior Probability

Marginal / evidence

$P(A|B)$ $\Rightarrow$ Probability of A given B has already occurred.
↑
given

Derivation of Bayes theorem

A: it is raining

B: going for shopping

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{--- (I)}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \quad \text{--- (II)}$$

From fundamental formula

Since $P(A \cap B) = P(B \cap A)$, [intersection is commutative]

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$\Rightarrow P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Q Given that, Probability that it rains on Saturday is 25%.
If it rains on Saturday, the probability it rains on Sunday is 50%.
If it doesn't rain on Saturday, the probability it rains on Sunday is 25%.
Given, that it rained on Sunday, what is the probability that it rained on Saturday.

Solution:

A: Rain on Saturday

$$P(A) = 0.25$$

B: Rain on Sunday

$$P(B|A) = 0.5$$

$$P(B|\bar{A}) = 0.25$$

$$P(A|B) = ?$$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Here, $P(B) = P(\text{rain on Sunday})$

rained on Saturday

didn't rain on Saturday

$$\text{So, } P(B) = P(A) \cdot P(B|A) + P(\bar{A}) \cdot P(B|\bar{A})$$

$$= 0.25 \times 0.5 + 0.75 \times 0.25$$

$$= 0.3125$$

$$\therefore P(A|B) = \frac{0.5 \times 0.25}{0.3125} = 0.4 = 40\%$$

$$\boxed{\frac{P(B|A) \cdot P(A)}{P(B)} = P(A|B)}$$

Supervised Classification Algorithm

Naive Bayes Algorithm

Problem Statement: Given X as an email content. Predict if the email is spam or non-spam.

Solution: X is new email,

$$P(Y=1|X) = \frac{P(X|Y=1) \cdot P(Y=1)}{P(X)}$$

$$P(Y=0|X) = \frac{P(X|Y=0) \cdot P(Y=0)}{P(X)}$$

$$P(X) = P(X|Y=1) \cdot P(Y=1) + P(X|Y=0) \cdot P(Y=0)$$

not necessary to calculate

Let 100 emails $\begin{cases} 60 \text{ spam (1)} \\ 40 \text{ non-spam (0)} \end{cases}$

Here, Prior Probability

$$P(Y=1) = \frac{60}{100} = 0.6$$

$$P(Y=0) = \frac{40}{100} = 0.4$$

Email Index	Y
0	1
1	1
2	0
...	...
99	1

Ex: $X_{\text{test}} = \text{get unlimited mobile data at 95\% discount price}$

0.8 - $P(X_{\text{test}}|Y=1)$: In how many spam emails, these words are occurring. Count the frequency of these words in spam emails.

0.2 - $P(X_{\text{test}}|Y=0)$: In how many non-spam emails, these words are occurring. Count the frequency of these words in non-spam emails.

$$\therefore P(Y=1|X) \propto \overset{0.8}{P(X|Y=1)} \cdot \overset{0.6}{P(Y=1)} \Rightarrow \boxed{0.48}$$

$$P(Y=0|X) \propto \underset{0.2}{P(X|Y=0)} \cdot \underset{0.4}{P(Y=0)} \Rightarrow 0.084$$

Spam

Likelihood Formula

$$P(X|Y=1) \Rightarrow P(\langle x_1, x_2, x_3, \dots, x_n \rangle | Y=1)$$

$$\Rightarrow P(x_1|Y=1) \cdot P(x_2|Y=1, x_1) \cdot P(x_3|Y=1, x_1, x_2) \dots P(x_n|Y=1, x_1, x_2, x_3, \dots, x_{n-1})$$

Naive Bayes Assumption

- features are independent.
- Conditional independent assumption.

Independent means,

$$P(A|B) = P(A)$$

Event A: Roll a die and getting ≥ 3

Event B: In coin toss getting head

$$\Rightarrow P(x_1|Y=1) \cdot P(x_2|Y=1) \cdot P(x_3|Y=1) \dots P(x_n|Y=1)$$

So,

$$P(X|Y=1) = \prod_{i=1}^n P(x_i|Y=1)$$

Similarly,

$$P(X|Y=0) = \prod_{i=1}^n P(x_i|Y=0)$$

$$P(Y=1|X) = \frac{1}{\sum_{i=0}^n P(x_i|Y=1) \cdot P(Y=1)}$$

Play Golf Dataset

	OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY GOLF
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No



Test Data

today = (Sunny, Hot, Normal, False)

Golf Example.

$$P(y=1 | x) = \prod_{i=1}^n P(x_i | y=1) \cdot P(y=1)$$

$\nearrow 9/14$

$$P(y=0 | x) = \prod_{i=1}^n P(x_i | y=0) \cdot P(y=0)$$

$\nearrow 5/14$

$$P(\text{Outlook} = \text{sunny} \mid \underbrace{Y=1}_{\substack{\text{given} \\ \text{happened}}}) \Rightarrow \frac{3}{9}$$

$$P(\text{Temp} = \text{Hot} \mid Y=1) = \frac{2}{9}$$

$$P(\text{Humidity} = \text{Normal} \mid Y=1) = \frac{6}{9}$$

$$P(\text{Wind} = \text{false} \mid Y=1) = \frac{6}{9}$$

$$P(\text{outlook} = \text{sunny} \mid Y=0) = \frac{2}{5}$$

$$P(\text{Temp} = \text{Hot} \mid Y=0) = \frac{2}{5}$$

$$P(\text{Humidity} = \text{Normal} \mid Y=0) = \frac{1}{5}$$

$$P(\text{Wind} = \text{false} \mid Y=0) = \frac{2}{5}$$

$$p(Y=1 | X) \Rightarrow \prod_{i=1}^n p(x_i | Y=1) \cdot \underbrace{p(Y=1)}$$

$$\Rightarrow \frac{3}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14}$$

$$\Rightarrow 0.0141$$

$$p(Y=0 | X) = \frac{2}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}$$

$$\Rightarrow 0.0068$$